21/9/24

Expt 9:

Implementation of Subnetting in Cisco packet tracer Simulator.

Classes IP Subnetting is a technique that allow for more efficient use of IP address by allowing Subnet mask that are not just default mask for each IP class creating a network topology

The first step in classes IP to create network topology in packet. No create a network topology, Select "New" button then Select "Network" "Generic" this will create an blank topology

Adding the device.

After creating topology we can add device we will add switch router. PC.

Subnetting:

To Subnet the network address 192.168.1.0/24 to provide atleast 5 address the end devices. The switch router, we can use a subnet mask the IP addressing for the network.

Router R1:

gigabit ethernet 0/0: 192.168.1.1

gigabit ethernet 0/1/1: 192.168.2.1

Switch S1:

FastEthernet 0/1: 192.168.1.0/27

PC1: 192.168.1.11

PC2: 192.168.1.12

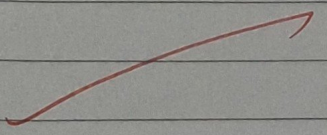
PC3: 192.168.1.13

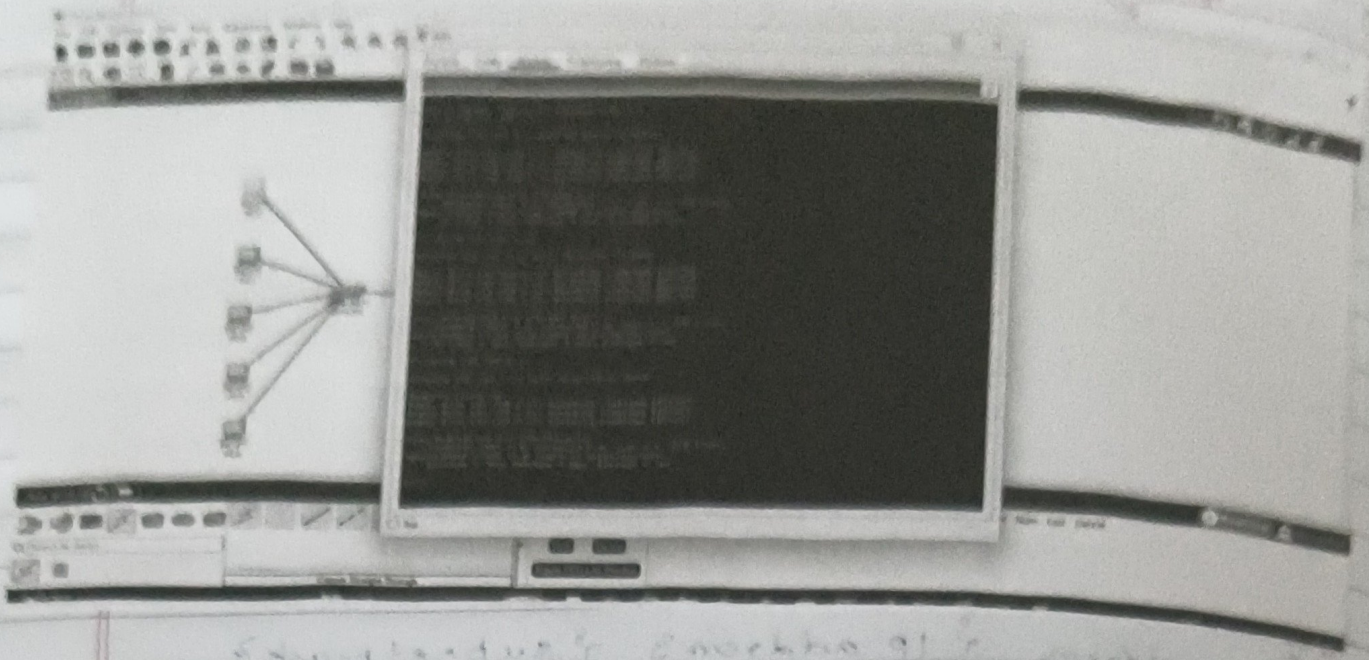
PC: 192.168.1.15

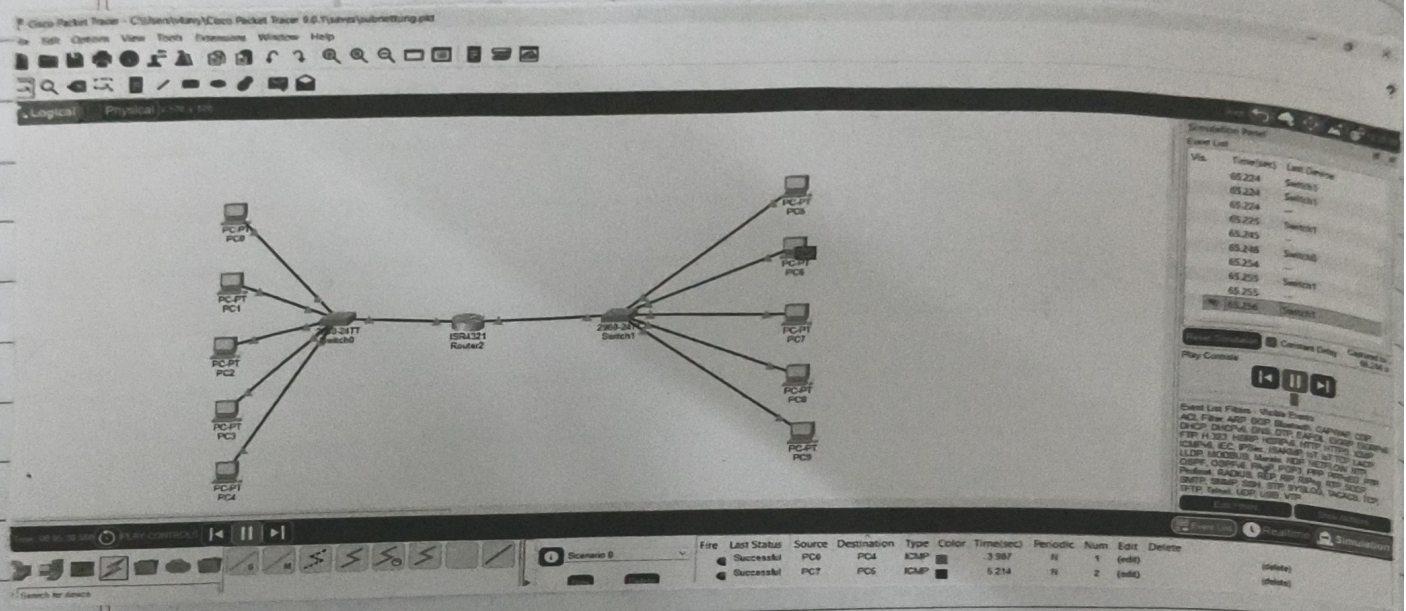
Configuring the devices.
After adding & Connecting the device, we can start configuring by typing commands in CLI

#enable
#Configure FastEthernet 0/0
#ip address {IP address} {subnet mask}
#no shutdown
#Exit.
interface FastEthernet 0/1
ip address {IP address} {subnet mask}
no shutdown
exit

Testing the network
open a command prompt on each pc
try to ping other pc if ping is successful then network is functioning properly.







Result:

Therefore implementation of Subnetting in cisco packet tracer has been executed successfully.